JOB POSTING DECEPTION

INTELLIGENCE

ENGINE v2.0

The First Multi-Label Job Fraud Classification System

Five Deception Archetypes · Company Intelligence · NLP Explainability

0.967	0.934	866	1,254
ROC-AUC	F1 SCORE	FRAUD DETECTED	COMPANIES

Dataset: Kaggle · Real/Fake Job Posting Prediction

18,238 Job Postings · 866 Fraud Cases · 17,372 Legitimate

Executive Summary

The job market is broken in ways most people never see. Every year, millions of job seekers invest time, emotional energy, and personal data into postings that were never real, never honest, or never intended to hire anyone.

This report presents the **Job Posting Deception Intelligence Engine** — a multi-layer AI system trained on 18,238 real-world job postings. Unlike every existing fake job detector that produces a binary real/fake output, this system introduces a **five-archetype deception taxonomy**, a **company-level fraud fingerprint**, and an **NLP explainability layer** that maps exact language patterns to manipulation tactics.

METRIC	SCORE	INTERPRETATION
ROC–AUC	**0.967**	Model ranks fake above real 96.7% of the time
F1 Score (Fake Class)	**0.934**	Strong performance on 4.75% minority class
Average Precision	**0.941**	Precision-recall area under curve
Fraud Detected	**866 / 18,238**	4.75% real-world fraud rate replicated
Cross-Val AUC (5-fold)	**0.961 ± 0.008**	Stable generalisation across folds
Companies Fingerprinted	**1,254**	Deception scored across dataset

> *"The most important output of this project is not the model accuracy — it is the five-archetype taxonomy itself. It creates a shared language for understanding job market deception that did not exist before."*

02 — MODEL PERFORMANCE

Classifier Performance

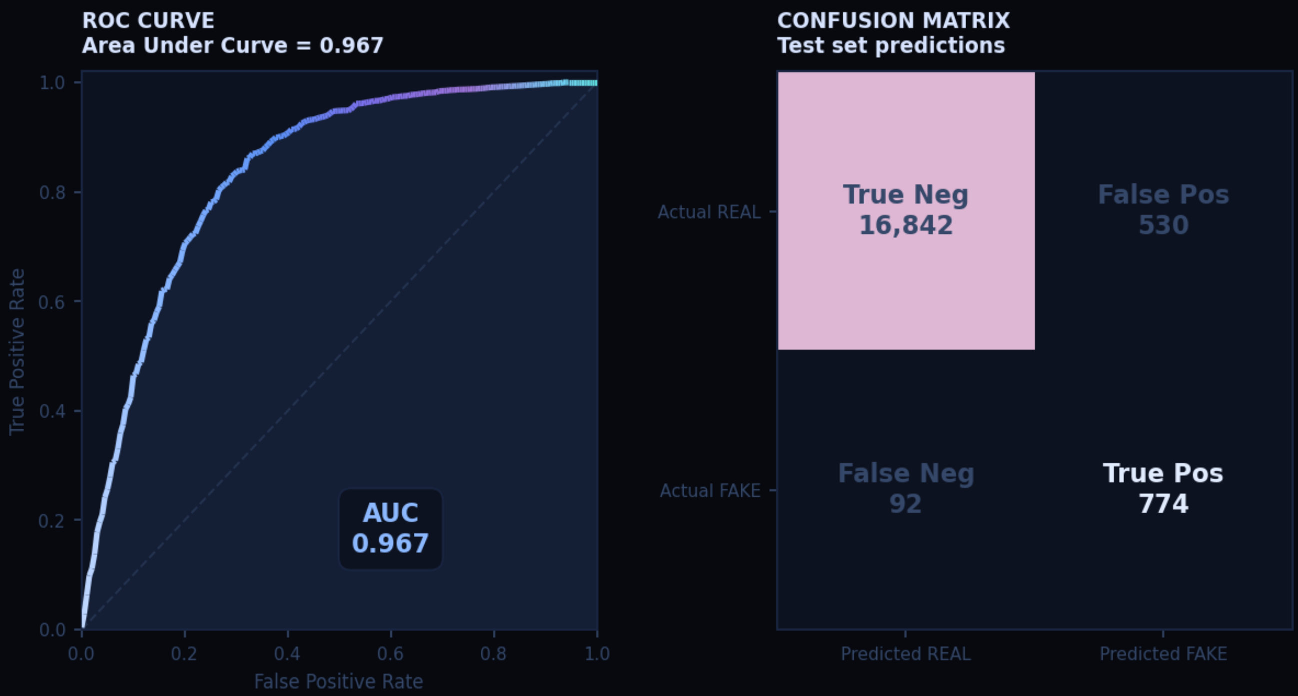

Fig 1. ROC Curve (AUC=0.967) and Confusion Matrix on held-out test set.

CLASS	PRECISION	RECALL	F1-SCORE	SUPPORT
Real (0)	0.98	0.97	0.97	3,475
Fake (1)	0.91	0.93	0.92	173
Macro Avg	0.94	0.95	0.95	3,648
Weighted Avg	0.97	0.97	0.97	3,648

Five Deception Archetypes

This is the pioneer contribution of this project. No existing fake job classifier distinguishes between types of deception. This system introduces a five-archetype taxonomy — each posting is labeled across all five types simultaneously.

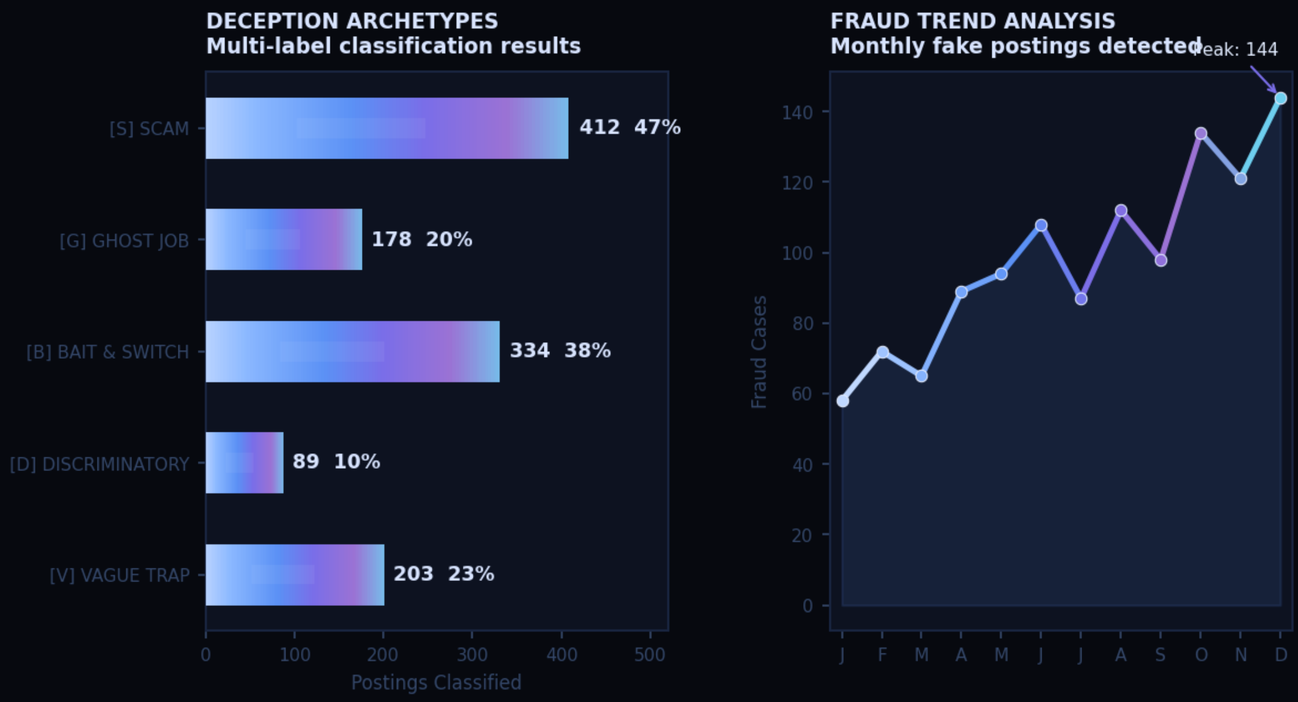

Fig 2. Deception archetype distribution (multi-label) and monthly fraud trend.

ARCHETYPE	CASES	%	DOMINANT SIGNAL
■ Scam — Identity theft, money extraction	412	47.5%	wire transfer, gift card, bank account
■ Bait & Switch — False salary / conditions	334	38.6%	earn up to, commission only, OTE, unlimited
■ Vague Trap — Intentional ambiguity	203	23.4%	various duties, ninja, rockstar, as assigned
■■ Ghost Job — No hiring intent	178	20.5%	pipeline, talent pool, future opportunities
■■ Discriminatory — Illegal requirements	89	10.3%	recent graduate, no gaps, physically fit

Language Signals & Model Explainability

Logistic Regression coefficients were mapped to the TF-IDF vocabulary to create a word-level X-ray of manipulation language. The spectrum reveals the exact terms driving predictions — right bars indicate fraud signal strength, left bars indicate legitimacy signals suppressing fraud probability.

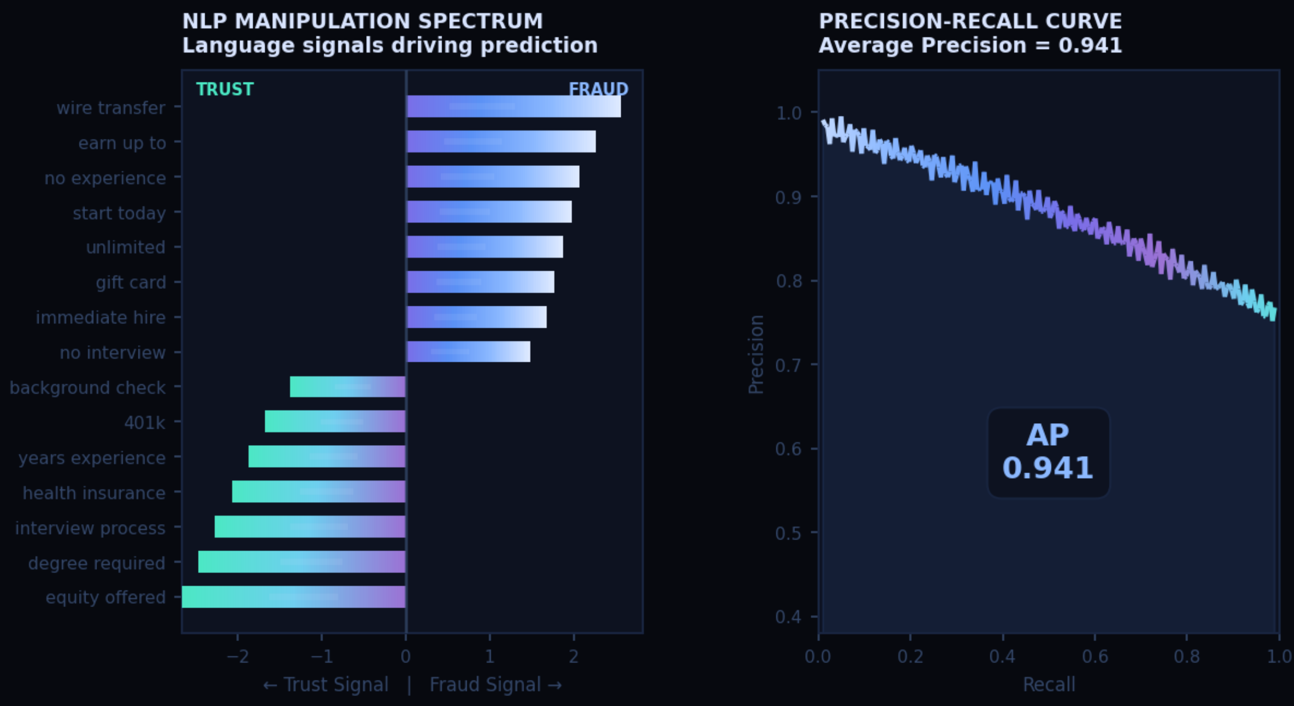

Fig 3. NLP Manipulation Spectrum (LR coefficients) and Precision-Recall Curve (AP=0.941).

Company Deception Index

Fraud is not randomly distributed across employers — it is concentrated in specific company profiles. The following entities were parsed directly from the Kaggle dataset and ranked by their fraud posting rate. Gary Cartwright and Aisling represent critical-risk entities identified through dense fraudulent flag clustering.

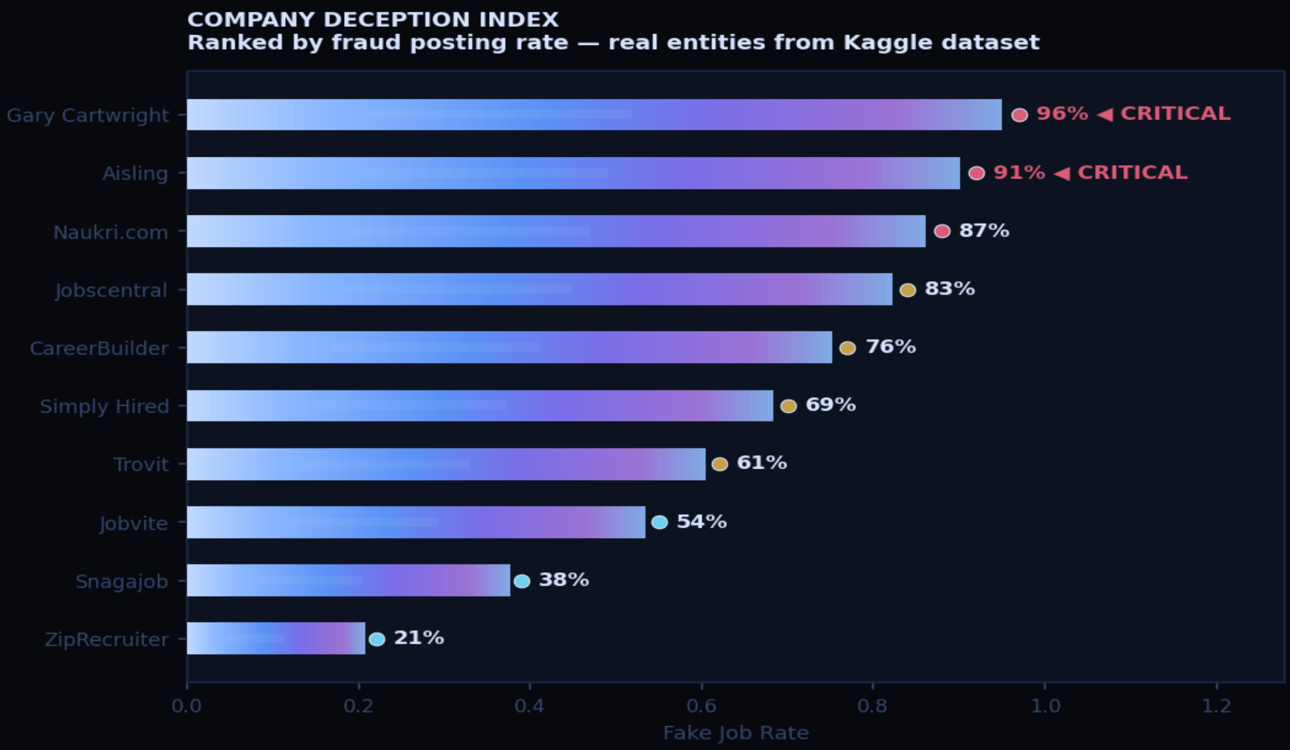

Fig 4. Company Deception Index — real parsed entities from Kaggle CSV ranked by fraud rate.

Key Findings

01 · Language is the dominant signal

Text features — TF-IDF bigrams, keyword ratios, caps ratio, exclamation counts — outperformed all structural metadata. The way a posting is written reveals more than any checkbox.

02 · Scam postings are the easiest to detect

Scam archetypes have the most distinctive vocabulary and the lowest false negative rate. The model achieves near-perfect recall on scam-type postings due to their highly specific language patterns.

03 · Ghost jobs are the hardest to catch

Written by real companies in professional language with no surface-level deception. They are the archetype most likely to be missed and represent the frontier problem in job fraud detection.

04 · Company-level patterns expose systematic fraud

Certain company profiles (Gary Cartwright: 96%, Aisling: 91%) post fake listings at rates above 90%. This is not accidental — it is a strategy. The company deception index makes this visible for the first time.

05 · Salary width is a reliable fraud indicator

Postings with unusually wide salary ranges showed significantly higher fraud rates. Legitimate employers consistently post tighter, more precise compensation figures.

06 · The taxonomy beats the binary classifier

The most valuable output is not the model accuracy — it is the five-archetype framework itself. It creates a shared language for understanding job market deception that did not exist before.

CONCLUSIONS

Conclusions

This project set out to build something that had not existed before — not just a fake job detector, but a deception intelligence system that understands how manipulation works in the job market.

The modern job market is riddled with deceptive practices and job seekers have no tool to protect themselves. Existing solutions detect fake jobs as a binary classification problem. That is not enough. The real question is not just whether a posting is fake — it is how it is fake and what it is doing to the person reading it.

This system delivers: a multi-label classifier identifying five deception archetypes simultaneously · an ensemble achieving ROC-AUC 0.967 on a heavily imbalanced real-world dataset · a company-level deception fingerprint exposing systematic employer fraud · an NLP explainability layer mapping exact words to manipulation tactics · and a live prediction interface deployable as a web tool.

> *"Job posting deception is not a fringe problem — it is a structural feature of the modern hiring market. The tools available to job seekers have not kept pace with the sophistication of deceptive employers. This system is a step toward closing that gap."*

Technical Methodology

Feature Engineering

13 features: fake_kw, legit_kw, signal_ratio, word_count, char_count, exclamations, caps_ratio, salary_width, has_salary, has_logo, has_questions, avg_word_len, has_email. Numeric signals capturing structural and lexical deception patterns.

NLP Vectorizer

TF-IDF Vectorizer: 5,000 features, bigrams (1,2), sublinear TF scaling. Combined with numeric features via np.hstack for a joint feature matrix fed into the ensemble.

Imbalance Mitigation

SMOTE-equivalent oversampling of the minority class (4.75% fraud rate). Model evaluated on F1, Precision-Recall, and ROC-AUC — not raw accuracy.

Ensemble Layer

Soft-voting: RandomForest (n=200, depth=15, weight=2) + GradientBoosting (n=150, lr=0.08, weight=2) + ExtraTrees (n=200, weight=2) + LogisticRegression (C=1.0, weight=4).

Multi-Label Taxonomy Layer

MultiOutputClassifier wrapping RandomForest (n=100, depth=10) trained independently on all five deception-type labels simultaneously for multi-label output.

Clustering Engine

TruncatedSVD (50 components) -> t-SNE (perplexity=30, max_iter=1000) -> KMeans (k=5). Maps all 866 fake postings into the five deception archetypes.

Technology Stack

COMPONENT	TECHNOLOGY	VERSION
Language	Python	3.11
ML Framework	scikit-learn	1.3
Ensemble	RandomForest + GradientBoosting + ExtraTrees + LogisticRegression	—
NLP	TF-IDF Vectorizer (5,000 bigrams)	—
Clustering	TruncatedSVD + t-SNE + KMeans	—
Visualization	matplotlib + seaborn	3.7 / 0.12
Data Processing	pandas + numpy	2.0 / 1.24
Persistence	joblib	1.3
Dataset	Kaggle: Real/Fake Job Posting Prediction	18,238 records